



Continuous Assessment Two

MACHINE LEARNING AND PATTERN RECOGNITION

Assessment:	2
Assignment Task	2
Grading Matrix	3
GenAI usage	5
Performance Descriptor Table	6
General Requirements for Students:	7

Assessment:

Item	Description
Module Title:	Machine Learning and Pattern Recognition
Module Code:	B9AI102
Module Lecturer (s):	Assem Abdel-Hak
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	Continuous Assessment
Percentage (%) Weighting:	60%
MIMLOs being assessed:	4,5,6
Assessment Number:	2
Individual/Group:	Group
Issue Date:	Week 8
Submission Date:	Week 12
Feedback Date:	Week 15
Feedback Strategy:	Moodle page

Assignment Task

Select a publicly available dataset with at least 10 input variables and 1500 instances to conduct statistical regression analysis in Python. [Obviously, you cannot select a problem/ dataset that has already been explored in the lectures] Implement linear regression and support vector regression techniques.

Deliverables

- 1- Dataset
 2. A PDF Report (named as Report_ Surname1_Surname2.pdf) **(60% weighting)**
 3. A Python Code File (named as Code_ Surname1_Surname2.py) **(40% weighting)**
- Submit both in a zipped folder named Surname1_Surname2.zip

There is no prescribed word count for the report; it will be assessed based on the quality of the content rather than quantity. Each group member should clearly define their individual contribution.
Your code and analysis should cover the following:

1. Chosen Problem and Data Preparation

(What steps would you take to prepare your data? Discuss your approach)

Key assessment areas

- Dataset selection/collection methodology
- Data preprocessing pipeline

2. Machine Learning Implementation & Model Hyperparameter Tuning

(Which hyperparameters would you tune and why? How would you tune them?)

Key assessment areas

- Implementation of different regression algorithms
- Hyperparameter optimization
- Performance metrics and evaluation
- Comparison of approaches

**3. Overfitting avoidance mechanism
(what you use and why)**

Key assessment areas

- Understanding of Overfitting
- Technique(s) Used
- Implementation Evidence

4. Performing a prediction for one of your models using new data.

Key assessment areas

- Creation or Selection of New Data
- Use of Trained Model for Prediction
- Clarity and Presentation

Grading Matrix

Criteria	Excellent (70–100%)	Good (60– 69%)	Satisfactory (50–59%)	Needs Improvement (40–49%)	Fail (0–39%)
1. Data Preparation & Dataset Selection [25%]	Dataset is well-chosen (not from class examples), rich (≥ 10 features, ≥ 1500 instances), clearly justified. Comprehensive preprocessing applied (cleaning, encoding, scaling, outlier handling, etc.).	Dataset is appropriate and mostly justified; preprocessing is good but lacks minor steps or rationale.	Dataset meets criteria but justification/preprocessing is limited or mechanical.	Dataset is only partially appropriate; preprocessing is weak or incomplete.	Poor dataset choice (too small, reused from class) and/or no meaningful data preparation.

2. Model Hyperparameter Tuning [25%]	Key hyperparameters (for Linear & SVR) are well-selected, justified, and tuned (e.g., GridSearchCV), with clear rationale.	Appropriate hyperparameters are tuned with partial explanation. Some insight shown.	Some tuning done but rationale is basic or lacks clarity.	Minimal effort, tuning poorly explained or ineffective.	No tuning attempted or only default parameters used with no justification.
3. Overfitting Avoidance [25%]	Strong awareness of overfitting; effective techniques used (e.g., cross-validation, regularization). Clear rationale provided.	Valid methods applied with some justification. Some insight into bias-variance trade-off.	Overfitting addressed at a basic level. Method lacks depth or clarity.	Weak handling of overfitting; unclear explanation.	No strategy to address overfitting; overfitted model produced.
4. Prediction with New Data [25%]	New input case is well-defined and realistic. Prediction shown and clearly interpreted. Demonstrates model generalization.	Prediction included and explained, but lacks depth or context.	Prediction provided with minimal interpretation.	Prediction unclear or underexplained.	No prediction or meaningless result shown.

GenAI usage

1	2	3	4	5
NO AI	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI-ASSISTED EDITING	AI TASK COMPLETION, HUMAN EVALUATION	FULL AI
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated

Performance Descriptor Table

		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)			
Scale	Grade Band	Criterion 1 – Dataset & Data Preparation	Criterion 2 – Hyperparameter Tuning	Criterion 3 – Overfitting Avoidance	Criterion 4 – Prediction
Exceptional	90–100%	Innovative, thorough preprocessing; dataset is rich, original, and perfectly justified.	Advanced tuning with original insight and optimal model configuration.	Expert use of advanced techniques (e.g., nested CV); strong theoretical grounding.	Realistic new input used; prediction process and result are insightful and well explained.
Outstanding	80–89%	Excellent preparation and thoughtful dataset justification.	Well-chosen parameters with justified tuning.	Clear, effective techniques used and well discussed.	Prediction task well executed and interpreted clearly.
Excellent	70–79%	Very good dataset and preparation with strong rationale.	Good tuning with some rationale.	Good application of standard techniques.	Prediction included and meaningfully interpreted.
Very Good	60–69%	Meets requirements, with decent preparation and a suitable dataset.	Tuning attempted, with some minor gaps.	Techniques applied, limited discussion.	Prediction shown, limited explanation.
Good	50–59%	Acceptable dataset and basic preprocessing.	Minimal or poorly explained tuning.	Weak or unclear strategy.	Prediction result shown, limited context.
Sufficient	40–49%	Weak dataset and basic preparation, lacks clarity.	Default settings mostly used.	Overfitting not adequately addressed.	Prediction vague or not well-executed.
Somewhat Unfulfilled	30–39%	Problematic dataset; barely any data prep.	Poor or no tuning.	Little or no awareness of overfitting.	Prediction missing or irrelevant.
Unfulfilled	0–29%	No real dataset or preparation effort.	No tuning.	No attempt to handle overfitting.	No prediction shown.
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)					
MIMLO to which criterion is aligned:		LO1, LO4	LO4	LO4, LO5	LO6

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
 - c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).
 - a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's Generative AI Guidelines HERE for further information.
 - c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
 - a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.

- d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.
 - e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)
6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.